



Psychological mechanisms of prenatal mindfulness training on antenatal blood pressure reduction: A pilot study[☆]

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ABSTRACT

Background: Hypertensive disorders of pregnancy (HDP) are a significant cause of perinatal morbidity and mortality. Mindfulness training (MT) shows promise for reducing blood pressure during pregnancy; however, the underlying mechanisms by which MT exerts this effect remain unclear.

Objectives: The objectives of this pilot randomized clinical trial were to study psychological mechanisms through which MT may reduce blood pressure in pregnancy.

Study design: Pregnant participants were randomized to an 8-week phone-delivered mindfulness intervention or usual care. Antenatal blood pressure values were extracted from medical records before and after the intervention to examine change in blood pressure. Participants also completed self-report measures of depression, anxiety, perceived stress, and mindfulness before and after the intervention.

Results: Twenty-nine participants were randomized to phone-based MT ($N = 15$) or usual care ($N = 14$). Blood pressure significantly decreased across participation among those randomized to MT vs. usual care (systolic blood pressure ($F = 5.093$, $p = .035$); diastolic blood pressure ($F = 9.689$, $p = .005$)). Reductions in perceived stress mediated the relationship between mindfulness training and decreases in systolic—but not diastolic—maternal blood pressure.

Conclusions: Findings from this pilot trial suggest that reductions in perceived stress may underlie the effect of prenatal mindfulness training on lowering maternal systolic blood pressure.

Clinical Trial Registration: The [ClinicalTrials.gov](https://clinicaltrials.gov) identifier is NCT03679117.

1. Introduction

Hypertensive disorders of pregnancy (HDP), a group of disorders including gestational hypertension, chronic hypertension, preeclampsia, and eclampsia, are a leading cause of maternal morbidity and mortality and impact 1 in 7 pregnancies in the United States [1]. Maternal and infant morbidity from HDP include placental abruption, severe neurological complications, cardiovascular disease, acute liver and kidney failure, fetal growth restriction, and sequelae of prematurity [2,3]. Pharmacological interventions to prevent HDP are used conservatively in pregnancy due to some perinatal risks [4–6], and non-

pharmacological interventions to prevent HDP, including exercise, vitamin intake, fish oil, and salt restriction, have not shown significant benefits on HDP risk [7,8]. Given potential fetal safety concerns and the fact that patients may be reluctant to take prescription medications in pregnancy [9], we examined prenatal mindfulness training as a non-pharmacological intervention to reduce risk for HDP.

Mindfulness training teaches participants to “pay attention in a particular way, on purpose, in the present moment and non-judgmentally, to the unfolding of experience moment by moment [10].” According to ‘Mindfulness Stress Buffering Theory’ [11], mindfulness training leads to improved health by reducing emotional

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reactivity to everyday stressors. As well, recent meta-analyses report that mindfulness training (MT) significantly reduces blood pressure in non-pregnant adults [12,13]. In pregnant participants at low risk for obstetric complications, higher trait mindfulness was associated with lower blood pressure [14] and improvements in parasympathetic function [15]. Our previous published findings from this pilot clinical trial showed that prenatal MT was associated with significant reductions in clinic-obtained systolic and diastolic blood pressure [16]. Now, we aimed to examine if improvements in psychological symptoms serve as a possible mechanism explaining blood pressure reduction.

The American Heart Association and European guidelines identify poor mental health as a risk factor for cardiovascular disease [17]. However, the risk factors included in the American College of Obstetricians and Gynecologists guidelines to assess clinical risk for preeclampsia [18] do not include maternal mental health. In a systematic review and meta-analysis of RCTs in patients with vascular disease, mindfulness-based interventions significantly reduced symptoms of stress, depression and anxiety [19], and simultaneously reduced blood pressure [19–21], however psychological changes were not examined as mechanisms leading to blood pressure improvements. Ecological momentary assessment studies in non-pregnant adults demonstrated that MT was associated with lower momentary stress [22,23], negative affect variability [24], and depression and anxiety symptoms [25], however these studies did not examine effects on blood pressure or other cardiovascular parameters. In a large clinical trial of an adapted mindfulness program (MT/BP) for adults with elevated BP, participants experienced reductions in BP, sedentary behavior, and dietary changes, along with increased mindfulness [24]. However, these outcomes—along with other psychological factors—were not evaluated as potential mechanisms of BP reduction. [26]. Qualitative evidence from the same study showed that MT resulted in greater self-awareness, attention control, and emotion regulation, which participants believed was beneficial to their cardiovascular health [27]. No studies to date, to our knowledge, have examined psychological mechanisms of MT on blood pressure or hypertension risk in pregnancy.

To address this important gap in the knowledgebase, in this pilot study we aimed to examine possible psychological mechanisms that may explain blood pressure reduction following prenatal MT. We hypothesized that participants randomized to MT would display greater reductions in psychological symptoms of stress, depression and anxiety, and would experience greater increases in mindfulness, relative to those randomized to usual care. We also expected that changes in psychological symptoms would mediate the association between MT and blood pressure reduction. This study will contribute new theoretical knowledge on maternal mental health as a risk factor for hypertension in pregnancy and will generate preliminary evidence on prenatal MT as a potential preventative option for reducing HDP risk by improving mental health.

2. Material and methods

Details of this pilot study have been previously published (see [16]). Briefly, participants were recruited from a large practice of approximately 22 prenatal care providers and > 1300 deliveries per year. Patients were identified through the electronic medical record and screened by phone to determine eligibility. Participants were eligible to participate if they were ≥ 18 years old, with a singleton pregnancy, English speaking, ≤ 20 weeks' gestation at enrollment, had a history of a hypertensive disorder (defined as chronic hypertension, gestational hypertension, and/or preeclampsia), and were not engaged in mind-body practices (yoga, mindfulness exercises (including on-line), or meditation more than once per week). Participants were randomized to MT (+ usual obstetric care) or usual care alone at a 1:1 ratio using the random number generator in Microsoft Excel. Randomization was not stratified by demographic characteristics such as maternal age or body mass index. Participants completed self-report measures before and after

their participation in MT/usual care. Measures of blood pressure before and after the intervention period were extracted from medical records. The study was approved by the institutional review board. All participants provided written informed consent prior to participation. The [ClinicalTrials.gov](https://clinicaltrials.gov) identifier is NCT03679117.

3. Experimental conditions

3.1. Mindfulness intervention (+ usual care)

Participants randomized to the MT condition received 30-min MT sessions by phone once a week for 8 weeks. The MT intervention was based upon the curriculum of the Mindfulness-Based Stress Reduction (MBSR) program [28] and was simplified to highlight two primary components: body scan and awareness of breath to cultivate attention to sensations in everyday experiences. The intervention was originally developed by our study team for patients with cardiovascular disease [29–31] and has previously been demonstrated to be feasible and acceptable in a perinatal population [16,32]. Each session included a didactic component in which the participant was taught and practiced a mindfulness exercise, the participant and instructor engaged in a discussion of at-home practice, and time was left for questions and answers. The intervention was delivered by a certified mindfulness instructor. Participants randomized to MT also received usual care as described below.

3.2. Usual care

Participants randomized to usual care continued to receive standard prenatal care, including anti-hypertensive medications and low dose aspirin, as appropriate. Medications were recorded at baseline and follow-up visits for all participants. Participants randomized to usual care were not discouraged from engaging in mindfulness-based interventions during the study; participation in mindfulness-based interventions outside of the study was measured at post-treatment. However, no participants randomized to usual care reported engaging in these activities.

4. Measures

4.1. Mindfulness

All participants completed the Five Facet Mindfulness Questionnaire (FFMQ) which measures the trait mindfulness in daily life on five subscales: observation, description, acting with awareness, non-judgmental awareness, and non-reactivity [33]. The FFMQ consists of 39 items, each rated on a 5-point Likert scale, ranging from 1 (never or very rarely true) to 5 (very often or always true). The total FFMQ score can be derived by summing the scores across all five dimensions (i.e., observation, description acting with awareness, non-judgmental awareness, and non-reactivity), providing an overall measure of mindfulness. Total scores range from 39 to 195 with higher scores indicating greater mindfulness. The FFMQ has been validated for use in pregnancy [34]. The total score has shown good convergent validity with symptoms of psychological well-being [35].

4.2. Perceived Stress Scale

The Perceived Stress Scale (PSS) is a 10-item measure which assesses the degree to which situations are appraised as stressful in the previous month. Participants rate each item on a Likert scale ranging from 0 (Never) to 4 (Very Often). Scores range from 0 to 40 with higher scores indicating greater perceived stress [36]. The PSS is considered valid for use in pregnant samples [37].

4.3. Edinburgh Postnatal Depression Scale (EPDS)

The EPDS is a 10-item measure of pre or postnatal depressive symptoms in the past 7 days [38]. Participants rate their depressive symptoms on a scale of 0 (not at all) to 3 (as much as I always could/most of the time). Scores range from 0 to 30 with higher scores reflecting greater symptoms of depression. This measure was developed to detect depression in the perinatal period and has demonstrated high sensitivity across studies [39].

4.4. Pregnancy-Specific Anxiety (PSA)

The PSA is a 13-item measure that examines the extent to which participants feel anxious, concerned, afraid, and panicky about their pregnancy in the past week [40]. Items regarding anxiety are embedded into a larger list of adjectives of pregnancy-specific positive and negative affective states. Each item is answered on a 5-point Likert scale from 0 (never) to 4 (always) with total scores ranging from 0 to 16. The four anxiety items are summed with higher scores indicating greater anxiety.

4.5. Blood pressure

Electronic medical records were reviewed to abstract information on in-office blood pressure readings before and after participation. Blood pressure was measured manually by prenatal care providers using a wall mounted aneroid sphygmomanometer, with the patient in a seated position.

5. Statistical analyses

Descriptive statistics were performed to characterize the sample. Analysis of covariance models were performed to explore differences in psychological symptoms by group at post-treatment, adjusting for symptoms at baseline. Scores on psychological measures were normally distributed as were maternal blood pressure values (skewness <2). Analyses followed the intent-to-treat principle for clinical trials. We then conducted linear regression analyses between 1) MT group and blood pressure, 2) MT and each measure of psychological distress, and 3) if measures of psychological distress partially explained the association between MT and blood pressure using the Baron and Kenny method for testing mediation (i.e., both MT group and measures of psychological distress at baseline and post-treatment, and blood pressure at baseline, entered as predictor variables, and blood pressure at post-treatment as the outcome variable) [41]. IBM SPSS Statistics 25 was used for statistical analyses.

Table 1
Participant characteristics.

	Mindfulness training N = 15	Usual care N = 14
	M (SD)/ N (%)	M (SD)/ N (%)
Maternal Age	33 (4)	32 (5)
White	11 % (73 %)	8 % (57 %)
Hispanic	2 % (13 %)	2 % (14 %)
Number of prior pregnancies	2 (1)	2 (1)
Body Mass Index (Kg/m ²)	30 (6)	28 (6)
Low dose Aspirin	13 % (87 %)	13 % (92 %)
Gestational age at baseline (weeks)	15 (3)	17 (3)
Gestational age at follow-up (weeks)	26 (5)	27 (4)
Baseline Hypertensive medications	2 % (13 %)	1 % (7 %)
Follow-up Hypertensive medications	0 % (0 %)	2 % (14 %)
Cigarette smoking	1 % (7 %)	1 % (7 %)
History of chronic hypertension	7 % (43 %)	6 % (43 %)

6. Results

Participant characteristics are shown in Table 1. Eighty-eight patients were approached at prenatal care appointments to determine interest in participation. Forty patients expressed interest and were screened for participation, and 30 participants enrolled in the study. Twenty-nine were randomized (one participant withdrew prior to randomization and was not included in subsequent analyses), with 15 assigned to MT and 14 to control group (See Fig. 1). Participants were, on average, 32 years old (SD = 4), 67 % White, 13 % Hispanic, body mass index was on average 29 (SD = 6). Forty-five percent of participants had chronic hypertension and 10 % were on anti-hypertensive medications at enrollment. Seventy percent of participants had a hypertensive disorder (gestational hypertension, preeclampsia, chronic hypertension) in a prior pregnancy. In addition, 90 % were prescribed daily low dose aspirin. Participants were, on average, 16 weeks' gestation (SD = 3) at enrollment and 26 weeks' gestation (SD = 5) at follow-up.

As reported in our previously published study [16], systolic blood pressure (SBP; $F = 5.093, p = .035$) and diastolic blood pressure (DBP; $F = 9.689, p = .005$) were significantly lower at post-treatment in participants randomized to MT compared with usual care. To test the current study hypotheses, we performed ANCOVA models to determine if, after adjusting for baseline values, psychological symptoms of mindfulness, stress, depression, or anxiety differed by MT group. Psychological symptoms did not differ by group at baseline (p -values > .10). Consistent with our hypothesis, symptoms of perceived stress were significantly lower among participants randomized to mindfulness compared to those in usual care, after adjusting for baseline perceived stress symptoms ($F = 6.916, p = .016$). In contrast to our hypotheses, there was no significant group difference in mindfulness ($F = 0.005, p = .945$), depressive symptoms ($F = 1.893, p = .188$), or pregnancy-specific anxiety ($F = 1.410, p = .248$) following the intervention period. See Table 2.

Finally, to examine if changes in maternal psychological symptoms mediated the association between MT and blood pressure, we performed linear regression. Study group, baseline blood pressure, and change in perceived stress between baseline and post-treatment were entered as predictors, and blood pressure measures at post-treatment were entered as outcome variables. Results revealed that, consistent with hypotheses, the association between study group and SBP after the intervention was no longer significant after adjusting for perceived stress symptoms ($B = -0.336, p = .228$), suggesting that the change in perceived stress fully accounted for the relationship between study group and SBP. The association between study group and DBP dropped to marginally significant after adjusting for perceived stress symptoms ($B = -0.392, p = .061$), suggesting that the change in perceived stress partly accounted for the relationship between study group and DBP.

7. Discussion

The goal of the current study was to examine whether prenatal MT reduces blood pressure during pregnancy among patients at risk for HDP through psychological mechanisms. Results from this pilot study indicated that perceived stress scores were significantly lower at post-treatment among those randomized to MT compared to usual care. As well, symptoms of perceived stress mediated the association between MT and SBP in this sample.

These findings are consistent with prior evidence linking MT to reductions in psychological stress. More specifically, a number of systematic reviews and meta-analyses have reported that mindfulness interventions are linked to decreased perceived stress in non-pregnant adults [42,43], including those with vascular disease [19], and in healthy pregnant adults [44]. While we did not replicate previous findings that MT was associated with reductions in depression and anxiety and increases in mindfulness [19,45], this was possibly due to

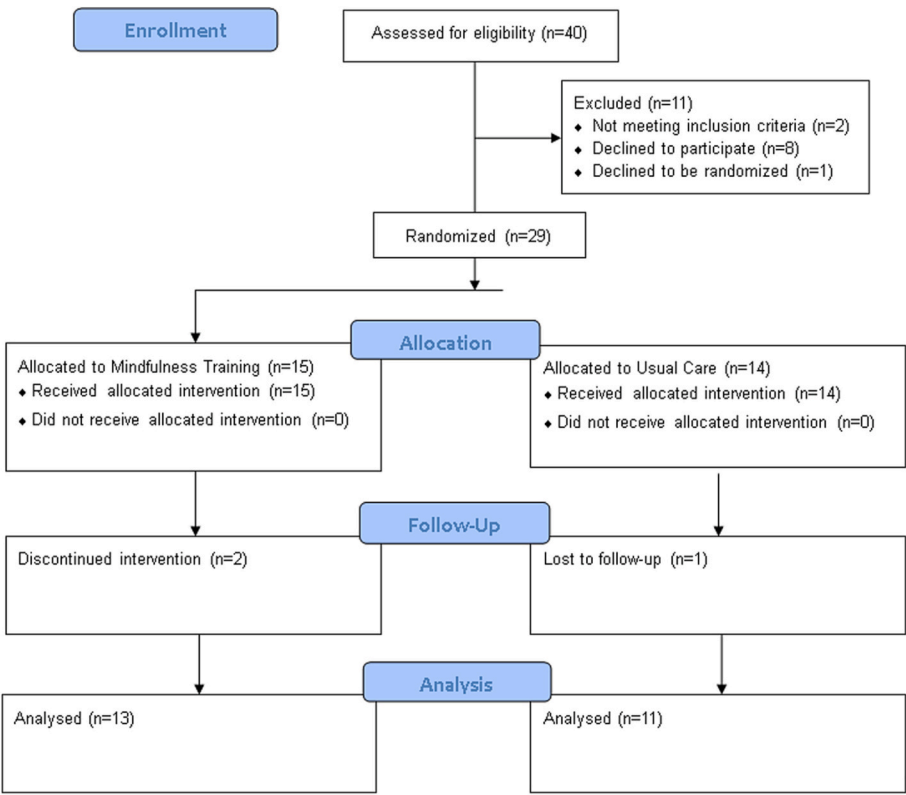


Fig. 1. CONSORT diagram.

Table 2
Psychological symptoms before and after the intervention.

	Mindfulness training N = 13	Usual care N = 11	F (p-value)
	M (SD)	M (SD)	
Perceived Stress Scale			
Pre-intervention	15.60 (4.79)	20.42 (8.60)	
Post-intervention	14.23 (5.07)	22.82 (7.57)	6.92 (0.02)
Edinburgh Postnatal Depression Scale			
Pre-intervention	3.27 (4.25)	5.54 (5.97)	
Post-intervention	3.75 (2.71)	6.00 (5.00)	1.89 (0.19)
Pregnancy Specific Anxiety			
Pre-intervention	3.72 (2.22)	5.21 (3.96)	
Post-intervention	3.75 (2.56)	5.25 (2.70)	1.41 (0.25)
Facets of Mindfulness Questionnaire			
Pre-intervention	102.53 (20.22)	90.15 (26.57)	
Post-intervention	103.54 (26.27)	90.54 (21.35)	1.89 (0.19)

Note. ANCOVA analyses adjusted for psychological symptoms at baseline.

our smaller sample size. As shown in Table 2, participants in the mindfulness group reported lower symptoms of depression and anxiety, and higher levels of mindfulness following the intervention relative to the usual care group, however these differences did not reach statistical significance. As well, we recruited participants based on medical risk factors for HDP rather than elevated psychological symptoms of distress. Therefore, the lack of significant findings may have been due to a “floor effect” such that participants were not experiencing clinically elevated

levels of stress, depression, or anxiety in their pregnancy. It is also possible that since we chose pregnancy-specific measures to assess anxiety and depression, that individuals' symptoms that were unrelated to pregnancy were not captured.

Findings indicated that perceived stress mediated the effects of prenatal MT on clinic-based SBP. This is in line with prior evidence that perceived stress is associated with SBP in young adulthood [46] and evidence that mindfulness-based interventions decrease both perceived stress and blood pressure [47–49]. As well, qualitative evidence suggests that mindfulness may improve blood pressure via increasing self-awareness, attention control, and emotional regulation [27]. Study results are consistent with the Mindfulness Stress Buffering Theory [11], proposing that MT results in reduced emotional reactivity in response to stress, and in turn improves health outcomes. Specifically, we observed that prenatal MT resulted in lower perceptions of stress, which mediated the association between MT and SBP. We are unaware of previous clinical trials in the pregnant population that have examined change in perceived stress as a mediator of the association between MT and blood pressure reduction. Therefore, this study adds new knowledge on the theoretical underpinnings of MT on health, and if replicated, has implications for clinical care, as results suggest that maternal perceived stress be considered as an additional risk factor for preeclampsia and HDP in American College of Obstetrician and Gynecological guidelines.

Decreases in perceived stress following MT were associated with reductions in SBP more so than DBP, consistent with prior studies [50,51]. Evidence from prior systematic reviews and meta-analyses of mindfulness-based interventions on blood pressure among adults with hypertension were mixed. Specifically, one study demonstrated that the benefits of MT on blood pressure reduction were particularly pronounced for DBP [52], while another demonstrated benefits for SBP only [53]. Taken together, this may suggest that psychological stress is not a mechanism linking MT and DBP reductions in pregnancy. More research is needed to explore additional mechanisms of MT, such as behavioral changes that have been reported previously, on DBP in pregnancy [26].

Results from this pilot study should be interpreted in the context of a number of limitations. First, the sample size was small and future larger trials are needed to replicate the findings. As well, blood pressure values were abstracted from medical charts and were collected as part of routine obstetric care. Future studies are needed that collect ambulatory blood pressure measures to assess daytime and nighttime blood pressure changes, including blood pressure dipping, as loss of dipping is a risk factor for the development of preeclampsia [54]. Moreover, participants may have underreported psychological symptoms in the perinatal period due to concerns of stigma [55]. Future studies could consider other methods of mental health symptom measurement, such as ecological momentary assessment, to increase ecological validity and minimize stigma concerns [56]. Another limitation is that we focused on very high risk population with 70 % having history of HDP in prior pregnancy. Future studies are needed to explore whether prenatal MT intervention can be effective in pregnant women at lower risk of HDP. Lastly, while groups appeared balanced on demographic characteristics that may impact maternal blood pressure, such as maternal age, future trials should stratify randomization to ensure balance across groups. Overall, the sample reported low levels of psychological distress. Future studies are needed to further test the Mindfulness Stress Buffering theory among participants with a higher prevalence of mental health conditions.

In conclusion, results from this pilot RCT suggest that prenatal MT may be associated with blood pressure reductions via decreases in perceived stress, particularly for SBP. Given the high prevalence of HDP and the limited number of interventions to prevent HDP onset, prenatal MT may be a promising adjunctive tool through which at-risk pregnant people can decrease blood pressure and mitigate HDP risk. Larger clinical trials are needed to confirm these findings and to explore additional mechanisms. Based on these findings, future studies that target pregnant people at risk for HDP and co-morbid elevated prenatal stress may be warranted.

CRedit authorship contribution statement

Margaret H. Bublitz: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Micheline Anderson:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Formal analysis. **Ghada Bourjeily:** Writing – review & editing, Writing – original draft, Resources, Methodology. **Elena Salmoirago-Blotcher:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis, Data curation.

Declaration of competing interest

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